

The Market Forces: Supply and Demand

Microeconomics

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Day 2

Where we are going today

- ① A puzzle to open
- ② Markets and competition
- ③ Demand
- ④ Supply
- ⑤ Equilibrium
- ⑥ Comparative statics
- ⑦ Robustness and pitfalls

October 1973: who quadrupled the oil price?

In a few months the world price of crude oil roughly **quadrupled**. Petrol queues formed from California to Yorkshire. No committee voted on the new price. No minister set it.

The question

Millions of buyers and thousands of sellers, none of them in charge, somehow converged on a new price within weeks. *What coordinated them?*

The answer is the most useful pair of curves in economics. By the end of today you will be able to take **any** such shock — an embargo, a frost, a viral trend, a new tax — and predict which way price and quantity move, and often by how much.

We return to the oil shock at the end, armed.

What exactly is a “market”?

Market

A group of buyers and sellers of a particular good or service. The market need not be a place — it can be a trading screen, an app, or a global network of phone calls.

Markets differ in how much competition they contain. Today we assume the cleanest case:

Perfect competition — two requirements

- ① The goods are (near) **identical**.
- ② Buyers and sellers are so **numerous** that no single one can influence the price.

Each participant is then a **price taker**: they decide only *how much* to buy or sell at the going price, not what the price *is*.

Monopoly, oligopoly and monopolistic competition come later — this is the benchmark they are measured against.

The modeller's scalpel: *ceteris paribus*

A market outcome depends on dozens of forces at once. To isolate one, we vary it and hold everything else fixed — *ceteris paribus*, “other things equal”.

- “How does quantity respond to **price**?” — holding income, tastes, other prices fixed.
- This is why a demand curve can be drawn in two dimensions at all.

Why it matters for the rest of the lecture

Almost every mistake in supply-and-demand analysis is a hidden violation of *ceteris paribus*: something the curve holds fixed has actually changed. Keep asking: “*what is being held constant here, and did the event break it?*”

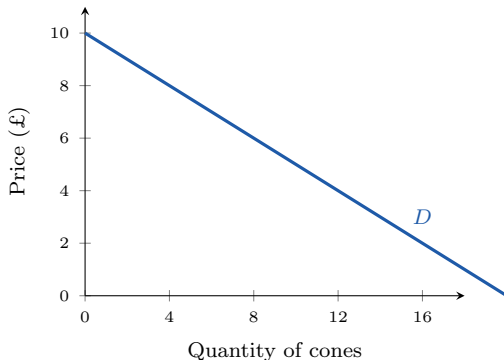
Quantity demanded and the demand schedule

Quantity demanded

The amount of a good buyers are *willing and able* to purchase at a given price. (Willing *and* able — a wish without purchasing power is not demand.)

A demand schedule for ice-cream cones:

Price (£)	Quantity
2	16
4	12
6	8
8	4
10	0



The schedule and the curve carry identical information. The curve slopes *down* — why, exactly? Three reasons.

Why does demand slope down? (Not “just because”)

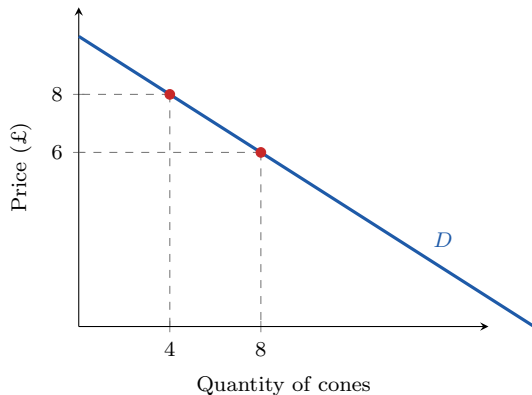
A fall in price raises quantity demanded through three distinct channels:

- ① **Substitution effect.** The good is now cheaper *relative* to alternatives, so buyers switch toward it. (Cheaper coffee → fewer teas.)
- ② **Income effect.** A lower price leaves more real spending power, some of which is spent on the good. (Cheaper petrol → you can afford to drive more.)
- ③ **New buyers enter.** People whose maximum price sat between the old and new price now buy for the first time.

Law of demand

Other things equal, a higher price reduces quantity demanded. For *normal* goods all three effects pull the same way — the curve slopes down robustly.

The demand curve is a *willingness-to-pay* curve



Read the curve *horizontally* as “how much at this price”.

Read it *vertically* as “what is the most the next buyer would pay”.

- The 4th cone is bought by someone valuing it at £8.
- The 8th cone’s buyer values it at £6.

Why this matters

The height of demand = marginal **value**. We will use this on Day 6 to measure the *gains* a market creates.

When the law of demand bends (robustness check)

The law of demand is empirically reliable — but not a theorem. Two textbook exceptions:

Giffen goods

A staple so dominant in a poor household's budget that when its price *rises*, the lost real income forces people to buy *more* of it (and less of pricier foods). The income effect overwhelms the substitution effect. Historically debated for potatoes; rigorously documented for rice and wheat among very poor households in field experiments.

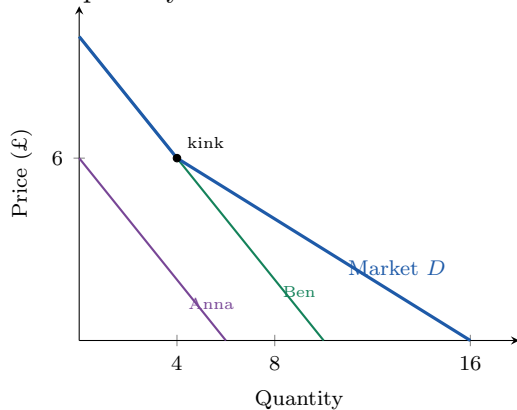
Veblen / conspicuous goods

For some luxuries a higher price *signals* status, raising desirability. Here “price” has entered the buyer's tastes — a hidden *ceteris paribus* violation.

The lesson is not “the curve is wrong” but “know the assumptions that make it slope down”.

From one buyer to the whole market

Market demand is the **horizontal sum** of individual demands: at each price, add up everyone's quantity.



Anna buys only when $P \leq 6$; Ben when $P \leq 10$.

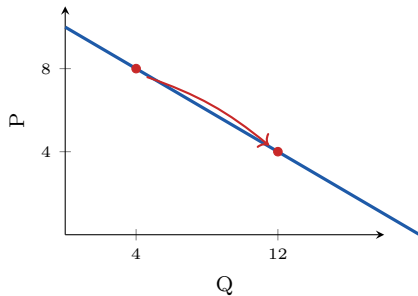
- Above £6: only Ben $\Rightarrow$ steeper.
- Below £6: both $\Rightarrow$ flatter (a **kink** where Anna enters).

More buyers, or buyers each wanting more, shift market demand *right*. This is why **number of buyers** is a demand shifter.

The distinction that trips everyone up

Change in *quantity demanded*

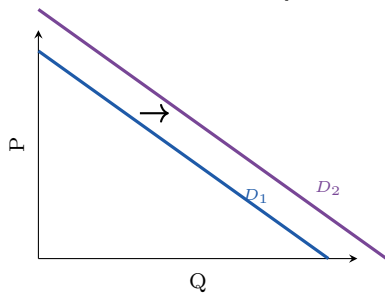
move *along* one curve



cause: the good's **own price**

Change in *demand*

the whole curve *shifts*



cause: **anything else** (income, tastes, ...)

Diagnose it

“A cold snap makes people buy fewer cones.” Movement or shift?

Shift (left): the cause is *weather*, not the cone price. Quantity falls at *every* price.

What shifts demand — with the direction made explicit

Shifter	Increases demand (right) when...	Watch out
Income	income $\uparrow$ <i>and</i> good is normal	for an inferior good (bus travel, instant noodles) the reverse
Price of substitutes	a substitute gets <i>dearer</i>	coffee $\leftrightarrow$ tea
Price of complements	a complement gets <i>cheaper</i>	printers $\leftrightarrow$ ink; cars $\leftrightarrow$ petrol
Tastes	the good becomes more fashionable	a viral trend, a health study
Expectations	buyers expect higher <i>future</i> prices	“buy now before it rises”
Number of buyers	the market grows	a new export market opens

Subtlety: “normal vs inferior” is *empirical* and can flip with income level — a good can be normal for the poor and inferior for the rich.

Your turn — demand diagnosis

Your turn

For the market for **cinema tickets**, state for each event: (i) shift or movement, and (ii) the direction.

- ① Streaming subscriptions get much cheaper.
- ② Average incomes rise (treat cinema as a normal good).
- ③ Cinemas cut ticket prices for a promotion.
- ④ A blockbuster everyone wants to see is released.

(1) Shift left — a cheaper substitute. (2) Shift right — normal good, income up. (3) Movement along (down the curve) — it's the ticket's own price. (4) Shift right — tastes.

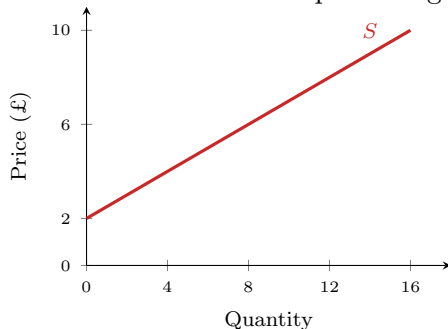
Quantity supplied — and why supply slopes *up*

Quantity supplied

The amount sellers are willing and able to sell at a given price.

Supply slopes up because of **rising marginal cost**: as a firm (or industry) expands output, it uses progressively less suitable resources — overtime wages, older machines, marginal land. A higher price is needed to make those extra units worth producing.

Price (£)	Quantity
2	0
4	4
6	8
8	12
10	16



Supply is a *marginal-cost* curve

Just as demand height is marginal value, **supply height is marginal cost**: the price on the curve at quantity Q is what it costs to make the Q -th unit.

- A seller supplies a unit whenever **price $\geq$ marginal cost**.
- Market supply is the horizontal sum of individual firms' supply.

One caveat for honesty

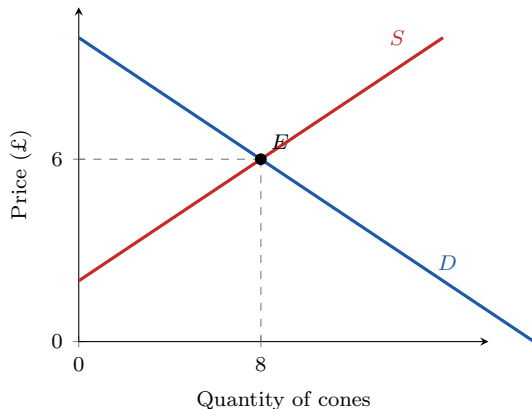
A firm's supply curve is its marginal-cost curve only *above* the point where price covers average variable cost; below that it shuts down rather than sell at a loss on every unit. We will make this precise in the theory of the firm.

What shifts supply — with the direction made explicit

Shifter	Increases supply (right) when...	Example
Input prices	inputs get <i>cheaper</i>	cheaper cream, lower wages, falling energy
Technology	a better production method arrives	automation, a higher-yield crop
Expectations	sellers expect <i>lower</i> future prices	sell now rather than store
Number of sellers	more firms enter	a new competitor opens
Nature / shocks	favourable weather, no disruption	a bumper harvest (right); a frost (left)

Same rule as demand: a change in the good's **own price** is a movement *along* supply; anything in this table *shifts* it.

Equilibrium: the market-clearing price



Equilibrium

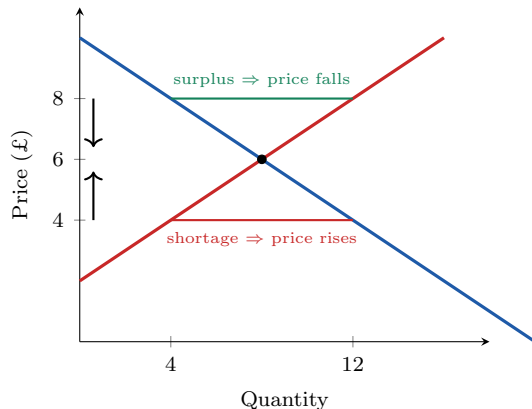
The price at which quantity supplied = quantity demanded; the market *clears*.

Solve $D = S$:

$$\begin{aligned} 10 - \frac{1}{2}Q &= 2 + \frac{1}{2}Q \\ 8 &= Q, \quad P^* = 6. \end{aligned}$$

At E every buyer willing to pay £6 is matched with a seller willing to accept £6. No one is rationed; no one is left holding stock.

Why the market reaches equilibrium



Above £6: a surplus piles up; sellers undercut to clear stock; price falls.

Below £6: a shortage appears; buyers bid up; price rises.

The only resting point is £6. Economists picture a fictitious “**auctioneer**” (Walras) calling prices until the market clears — a story for the real, messy process of haggling, restocking and re-pricing that does the same job.

Two confusions worth killing now

Scarcity $\neq$ shortage

Scarcity is the permanent condition that wants exceed resources — it never goes away. A *shortage* is a specific, temporary state where $Q_d > Q_s$ at the current price, cured by a price rise. Equilibrium has no shortage but is still scarce.

A high price $\neq$ a rising price

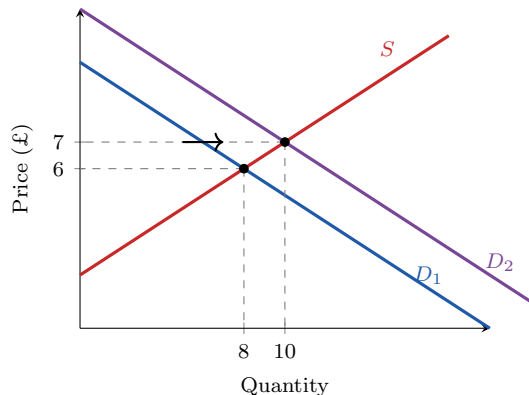
Equilibrium can sit at a permanently high price with *no* shortage (prime London property). Shortages come from prices that are *prevented from adjusting* — which is exactly what price controls do (Day 5).

The three-step recipe for any shock

- ❶ **Which curve?** Does the event shift supply, demand, or both?
(Test: is it the good's *own price*? Then neither — it's a movement.)
- ❷ **Which way?** Right (increase) or left (decrease)?
- ❸ **Compare equilibria.** Read off the change in P^* and Q^* .

We now run the recipe three times: a clean demand shift, a clean supply shift, and the hard case where *both* move.

Run 1 — a heatwave (demand $\uparrow$)



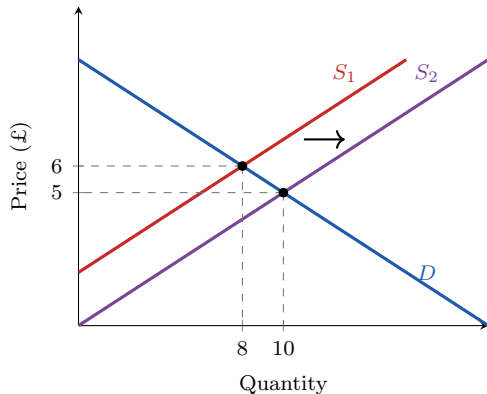
1. Hot weather changes *tastes* $\Rightarrow$ **demand**.
2. Right ($D_1 \rightarrow D_2$).
3. $Q^* : 8 \rightarrow 10$, $P^* : £6 \rightarrow £7$.

Algebra: $2 + \frac{1}{2}Q = 12 - \frac{1}{2}Q \Rightarrow Q = 10$, $P = 7$.

Result

Demand up $\Rightarrow$ *both* price and quantity rise.

Run 2 — cheaper inputs or better tech (supply $\uparrow$)



1. Lower cost / better tech $\Rightarrow$ **supply**.

2. Right ($S_1 \rightarrow S_2$).

3. $Q^* : 8 \rightarrow 10$, $P^* : £6 \rightarrow £5$.

Algebra: $\frac{1}{2}Q = 10 - \frac{1}{2}Q \Rightarrow Q = 10$, $P = 5$.

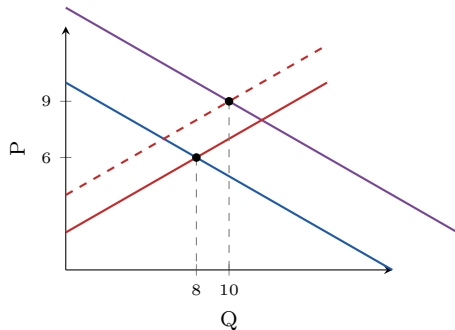
Result

Supply up $\Rightarrow$ price *falls*, quantity *rises*.

Run 3 — when *both* curves move, one outcome is ambiguous

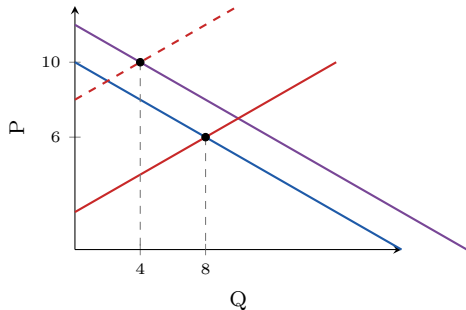
Demand shift dominates

$\Rightarrow Q \uparrow$



Supply shift dominates

$\Rightarrow Q \downarrow$



Both panels: demand *up* and supply *down* $\Rightarrow P^*$ definitely rises. Whether Q^* rises or falls depends on **which shift is larger** — the sign is genuinely undetermined without magnitudes.

The full ambiguity map

	Demand $\uparrow$	Demand $\downarrow$
Supply $\uparrow$	P ambiguous, $Q \uparrow$	$P \downarrow$, Q ambiguous
Supply $\downarrow$	$P \uparrow$, Q ambiguous	P ambiguous, $Q \downarrow$

- When the curves move the *same* way (Q): quantity is clear, price is ambiguous.
- When they move *opposite* ways (P): price is clear, quantity is ambiguous.

To pin down the ambiguous one you need the relative **sizes** of the two shifts — which is ultimately an *elasticity* question (Day 4).

Worked case — the 2022–23 egg-price spike

Two things hit the egg market at once: avian influenza forced the culling of tens of millions of laying hens (**supply left**), while seasonal baking demand peaked (**demand right**). Retail egg prices in several countries *more than doubled*, while quantity *fell*.

Why that combination $\Rightarrow$ price $\uparrow$, quantity $\downarrow$

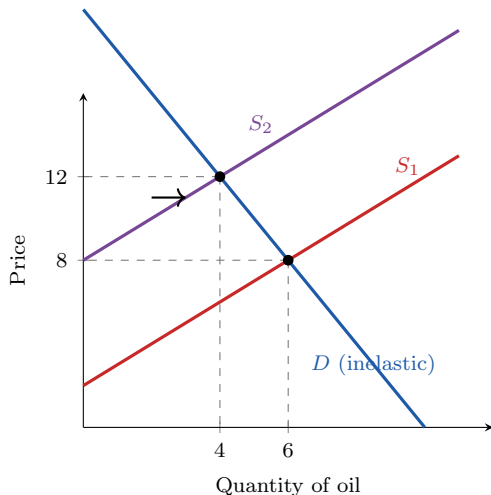
Demand up and supply down both push price up, so P^* rises unambiguously. The flock losses were large, so the *supply* shift dominated $\Rightarrow Q^*$ fell.

A stylised version, holding our slopes:

$$\underbrace{12 - \frac{1}{2}Q}_{\text{demand } \uparrow} = \underbrace{8 + \frac{1}{2}Q}_{\text{supply } \downarrow} \Rightarrow Q^* = 4, \quad P^* = £10.$$

Quantity $8 \rightarrow 4$, price $£6 \rightarrow £10$ — exactly the supply-dominates panel.

Back to the puzzle — the 1973 oil shock



1. The OPEC embargo cut the oil available $\Rightarrow$ **supply left**.
2. Demand is *steep* (inelastic): in the short run you still must drive and heat.
3. Result: price jumps a lot, quantity falls only a little.

The payoff

No one “set” the new price. The leftward supply shift against inelastic demand did — and the steepness of demand is exactly why the jump was so violent. *That* is what coordinated millions of buyers in weeks.

Common misconceptions & exam traps

- “Price went up, so demand rose.” Not necessarily — a *leftward supply* shift also raises price while *cutting* quantity. Always check quantity too.
- Confusing a movement with a shift. If the cause is the good’s own price, you move *along*; never redraw the curve.
- Shifting the wrong curve. A change in input costs moves *supply*, not demand.
- Declaring an outcome certain when both curves move. One of P, Q is ambiguous — say so, and say what would resolve it.
- Calling equilibrium “fair”. It clears the market; whether the result is *just* is a separate question (Day 6).

Where this model is strong — and where it breaks

The supply-and-demand model is astonishingly powerful, but it leans on assumptions:

- **Many price-taking participants.** Fails under *monopoly* / *oligopoly* — a seller who sets price.
- **No spillovers.** Fails with *externalities* — pollution, congestion — where private and social value diverge.
- **Good information & defined property rights.** Fails for *public goods* and where buyers can't judge quality.

The honest summary

Use supply and demand as the **default**. When a prediction feels wrong, the culprit is almost always one of these broken assumptions — and each is a named topic later in the course.

Key takeaways

- Demand slopes **down** (substitution + income + new buyers); supply slopes **up** (rising marginal cost).
- Curve *height* has meaning: demand = marginal value, supply = marginal cost.
- Own-price change = movement *along*; any other determinant = a *shift*.
- **Equilibrium** clears the market; surpluses and shortages drive price back to it.
- Analyse shocks in three steps; when both curves move, one of P^* , Q^* is ambiguous.

Coming next

Elasticity: how much quantity responds the number that decides the size of the oil-price jump, who pays a tax, and whether a good harvest helps farmers.

Questions?

EC107: Microeconomics